OXFORD

INTERNATIONAL
AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____
I declare this is my own work.

INTERNATIONAL AS
FURTHER MATHEMATICS

(9665/FM01) Unit FP1 Pure Mathematics

Tuesday 4 January 2022 07:00 GMT Time allowed: 1 hour 30 minutes

Materials
- For this paper you must have the Oxford International AQA Booklet of Formulae and Statistical Tables (enclosed).
- You may use a graphical calculator.

Instructions
- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.

Advice
- Unless stated otherwise, you may quote formulae, without proof, from the booklet.
- Show all necessary working; otherwise marks may be lost.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	

Do not write outside the box

Answer **all** questions in the spaces provided.

1 Show that

$$\sum_{r=n+1}^{2n} r^3 = \frac{1}{4}n^2(an+b)(bn+1)$$

where a and b are integers.

[5 marks]

5

Do not write outside the box

2 (a) It is given that $z = \dfrac{7-3\mathrm{i}}{k-5\mathrm{i}}$ where k is a real number.

Find, in terms of k, the real part of z and the imaginary part of z

[3 marks]

Re $z =$ _____ Im $z =$ _____

2 (b) Use your answer to **part (a)** to show that $\arg\left(\dfrac{7-3\mathrm{i}}{2-5\mathrm{i}}\right) = \dfrac{\pi}{4}$

[3 marks]

6

Do not write outside the box

3 A circular patch of water plants is floating on the surface of a lake.

The area of the circular patch increases at a rate of 3 square metres per day.

Find the rate at which the **radius** of the circular patch is increasing when the area of the circular patch is 36π square metres.

Give your answer in terms of π

[6 marks]

Answer _____

6

Do not write outside the box

4 (a) Find the general solution of the equation

$$\cos\left(2x - \frac{\pi}{2}\right) = -\frac{1}{2}$$

giving your answer in terms of π

[4 marks]

Answer

4 (b) Hence find the number of solutions of the equation

$$\cos\left(2x - \frac{\pi}{2}\right) = -\frac{1}{2}$$

which are between 0 and $(4k-1)\frac{\pi}{2}$ where k is an integer and $k \geq 1$

Give your answer in terms of k

[3 marks]

Answer

7

Do not write outside the box

5 The quadratic equation

$$x^2 + 5x + 9 = 0$$

has roots α and β

5 (a) Write down the value of $\alpha + \beta$ and the value of $\alpha\beta$

[2 marks]

$\alpha + \beta =$ _____ $\alpha\beta =$ _____

5 (b) Hence find the value of $\alpha^2 + \beta^2$

[2 marks]

Answer _____

5 (c) Show that $\alpha^3 + \beta^3 = 10$

[2 marks]

Do not write outside the box

5 (d) Find a quadratic equation, with integer coefficients, which has roots

$$\alpha + \frac{\beta}{\alpha} \quad \text{and} \quad \beta + \frac{\alpha}{\beta}$$

[6 marks]

Answer

12

Do not write outside the box

6 The function f is defined by

$$\mathrm{f}(x) = \frac{2x+1}{x^2}$$

6 (a) Write down the equations of the asymptotes of the graph of $y = \mathrm{f}(x)$

[2 marks]

Answer

6 (b) It is given that the line $y = k$, where k is a constant, intersects the graph of $y = \mathrm{f}(x)$

Prove that $k \geq -1$

[3 marks]

6 (c) The graph of $y = \mathrm{f}(x)$ has one stationary point.

Use the result given in **part (b)** to find the coordinates of this stationary point.

[2 marks]

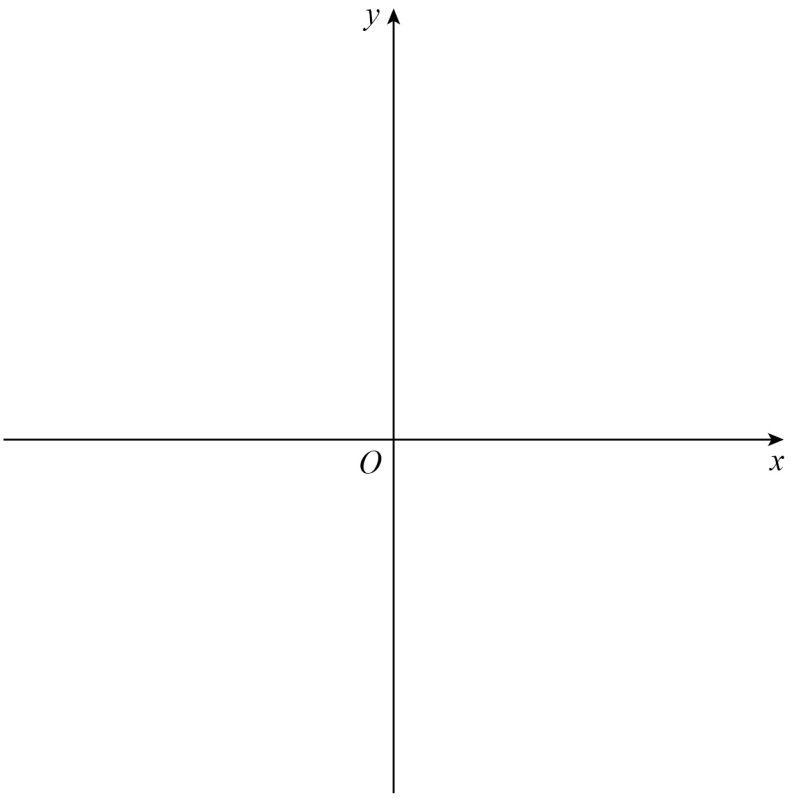
Answer

Do not write outside the box

6 (d) Sketch the graph of $y = \mathrm{f}(x)$ on the axes below.

Show the coordinates of the stationary point and of any intersection with the axes.

[3 marks]



6 (e) Solve the inequality

$$\frac{2x+1}{x^2} > 3$$

[4 marks]

14

Do not write outside the box

7 The integrals I_1 and I_2 are defined below.

$$I_1 = \int_1^{\infty} \frac{1}{x^2}\,\mathrm{d}x \qquad \text{and} \qquad I_2 = \int_0^{64} \frac{1}{\left(\sqrt[3]{x}\right)^2}\,\mathrm{d}x$$

7 (a) Explain why I_1 is an improper integral.

[1 mark]

7 (b) Explain why I_2 is an improper integral.

[1 mark]

Do not write outside the box

7 (c) Find the value of I_2 clearly showing the limiting process.

[3 marks]

Answer _____

5

Turn over for the next question

Do not write outside the box

8 The circle C is the locus of points on an Argand diagram such that

$$|z - 4 - 6\mathrm{i}| = 4$$

The half-line L is the locus of points on an Argand diagram such that

$$\arg(z - 4) = \frac{\pi}{2}$$

P is the point on C with the smallest argument.

The line OP meets the half-line L at the point T

8 (a) Write down the complex number that represents the centre of C

[1 mark]

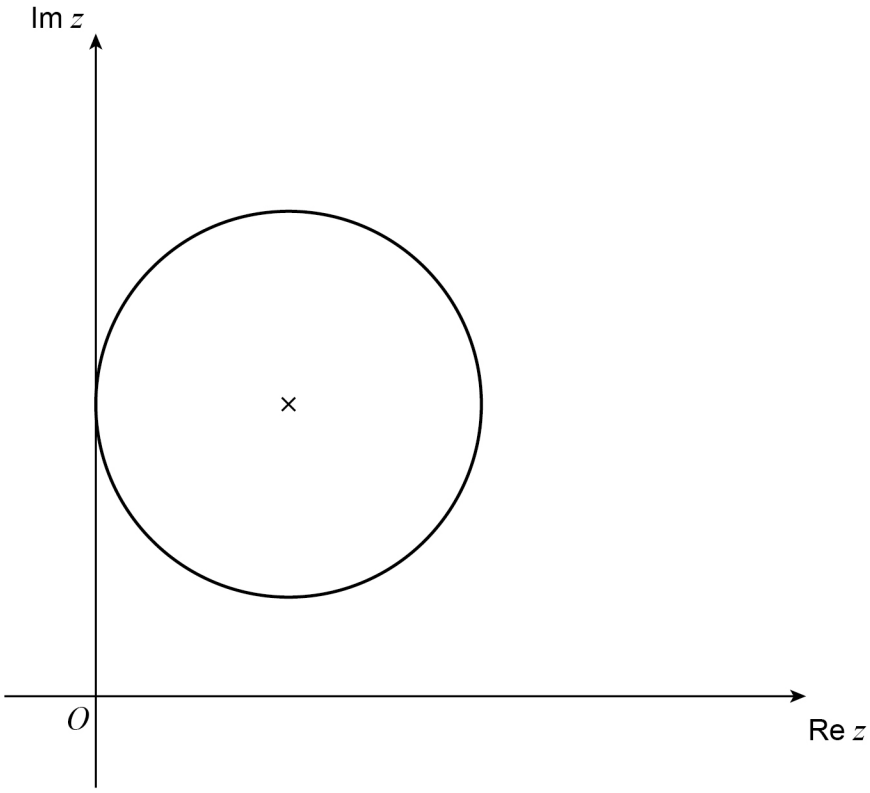
Answer _____

8 (b) The circle C is drawn on the Argand diagram below.

On the Argand diagram:

- draw the half-line L
- draw the line OP
- label the point T

[3 marks]



Do not write outside the box

8 (c) Find the complex number that represents the point T

[6 marks]

Answer

10

Do not write outside the box

There are no questions printed on this page

DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

Do not write outside the box

9 The locus of a point P is such that the distance from P to the point (12, 0) is equal to the distance from P to the line $x = -12$

The locus of P is the curve C_1

9 (a) Show that the equation of C_1 is

$$y^2 = 48x$$

[2 marks]

9 (b) The translation by the vector $\begin{bmatrix} 5 \\ 4 \end{bmatrix}$ maps the curve C_1 onto the curve C_2

Find the equation of C_2

[2 marks]

Answer _____

Question 9 continues on the next page

Do not write outside the box

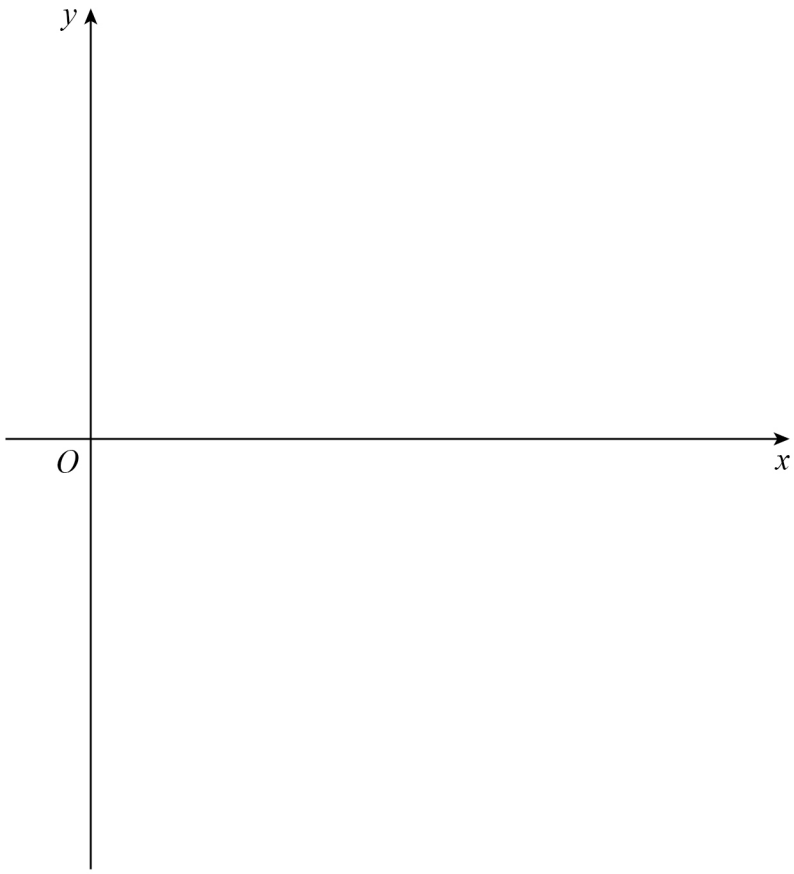
9 (c) The points Q and R both lie on C_2 such that

- the y-coordinate of Q is positive
- the y-coordinate of R is negative
- the line OQ is a tangent to C_2
- the line OR is a tangent to C_2

9 (c) (i) Sketch the curve C_2 and the lines OQ and OR

[You do **not** need to sketch the curve C_1]

[3 marks]



Do not write outside the box

9 (c) (ii) Find the coordinates of Q and the coordinates of R

[8 marks]

Q _____ R _____

END OF QUESTIONS

15

Do not write outside the box

There are no questions printed on this page

DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

Do not write outside the box

Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

Do not write outside the box

Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

Copyright information

For confidentiality purposes, all acknowledgements of third-party copyright material are published in a separate booklet. This booklet is published after each live examination series and is available for free download from www.oxfordaqaexams.org.uk.

Permission to reproduce all copyright material has been applied for. In some cases, efforts to contact copyright-holders may have been unsuccessful and Oxford International AQA Examinations will be happy to rectify any omissions of acknowledgements. If you have any queries please contact the Copyright Team.

Copyright © 2022 Oxford International AQA Examinations and its licensors. All rights reserved.